

Velocity Degradation in Hierarchical Multi-Agent Systems Under Discard

Abstract

Multi-agent coordination architectures are increasingly marketed as enabling scalable, parallel productivity through hierarchical orchestration. This paper analyzes a structural constraint on effective system velocity within a specific architecture class: hierarchical orchestration with partial output discard under uncertainty.

Within a formally specified model interface, this paper models net velocity for an n -agent hierarchy with branching factor k , per-agent output capacity c , expected discard burden m , and convergence tolerance ε as

$$v(n) = cn - kn \log_k(n) - mn\varepsilon^{-4}.$$

This paper proves that the coordination drag term $kn \log_k(n)$ is a structural lower bound under bounded fan-in hierarchies (Theorem 5.1), and models the discard term $mn\varepsilon^{-4}$ as a modeled scaling regime within the stated interface (Model 5.2). Under the proved coordination drag bound (Theorem 5.1) combined with the modeled discard regime (Model 5.2), there exists a threshold n_0 beyond which $v(n)$ decreases monotonically as n increases.

This paper presents a proof for the coordination drag component and a model justification for the discard component. Assumptions include: (1) hierarchical merge structure with branching factor k , (2) partial discard at merge interfaces, (3) uncertainty governed by convergence tolerance ε , (4) per-agent output capacity c per coordination cycle, and (5) discard/evaluation burden m proportional to agent count.

The result is derived within the model interface and does not generalize beyond these assumptions. It is offered as a diagnostic constraint rather than a prescription or evaluation of any specific implementation.

Keywords: - distributed systems - multi-agent coordination - hierarchical orchestration - coordination overhead - merge cost - discard - pruning - scaling limits - conditional theorem - threshold behavior JEL CODES: - C60 - C63

0. Posture and Admissibility Declaration

This paper is:

- Diagnostic, not interventionist
- Structural, not behavioral
- Non-prescriptive, offering no remedies or best practices
- Non-predictive, making no forecasts
- Architecture-class scoped, without claims of universality

All claims are admissible only within the interface explicitly declared below.

Guardrail A: This document is descriptive (diagnostic), not a basis for directing or evaluating human conduct. **Guardrail B:** Any reading that treats “restraint” as virtue, mandate, or moral ranking is outside the scope of this analysis. **Guardrail C:** Nothing here defines “resolution” as KPIs, throughput, or optimization targets; those substitutions change the object under study.

1. Scope and Boundaries

1.1 In Scope

- Hierarchical orchestration systems in which an orchestrator (or stack of orchestrators) repeatedly **merges** subordinate outputs and **discards** some portion to maintain coherence under uncertainty
- A structural notion of **effective velocity**: progress through task-space toward a coherent completion state (distinct from raw throughput)
- Coordination drag and discard/evaluation burden as scaling terms
- A conditional theorem about threshold behavior of $(v(n))$ within the model interface

1.2 Out of Scope

- Individual competence, motivation, or “productivity” of humans or teams
- Leadership quality, managerial judgment, or blame attribution
- Optimization advice, best practices, architectural prescriptions, or policy recommendations
- Product-level evaluations or claims about any specific vendor system
- Forecasting future technology trajectories

1.3 Interpretive Boundary

Any interpretation that imports optimization goals, performance norms, or authority to evaluate human conduct from outside this model interface is non-admissible.

2. Problem Statement

Hierarchical orchestration is frequently presented as a scaling path: add specialized agents, coordinate them through a lead orchestrator, and obtain more usable progress. Yet architectures that must repeatedly evaluate, merge, and discard outputs to preserve coherence under uncertainty face a structural cost: coordination and discard burden may grow faster than output contribution as the hierarchy deepens.

This paper isolates that architecture class and asks a narrow question:

How does effective velocity scale as the number of agents increases when coherence is maintained by hierarchical merging with partial discard under uncertainty?

3. Definitions

3.1 Effective Velocity (not throughput)

In this paper, **velocity** is not speed, utilization, or output rate. It denotes the rate at which **meaning-bearing change** propagates through the coordination system toward a coherent completion state.

A system can show high activity (many tokens, many tasks, many merges) while exhibiting low velocity (little coherent progress). This distinction is the object of study.

The concept is referred to as effective velocity and formalized as net velocity $v(n)$, the algebraic sum of output and drag terms.

3.2 Discard at Merge Interfaces

At a merge interface, an orchestrator receives multiple candidate outputs and must select and integrate a subset. The remainder is discarded (not propagated upward). Discard is treated as a structural operation with cost, not as moral failure.

3.3 Vestigium (optional term; descriptive)

Discard can create a **legibility remainder**: exploration outputs that do not propagate upward as legible constraint information. Where useful, this remainder may be referred to as *vestigium*—purely descriptive, with no implied evaluation.

4. Model Specification

A hierarchical multi-agent coordination system is defined by the tuple:

$$(n, k, c, m, \varepsilon)$$

- n : number of leaf-level agents producing candidate work per coordination cycle
- k : branching factor (fan-in) for merges (each merge integrates k subordinate outputs)
- c : per-agent output capacity per coordination cycle
- m : expected discard/evaluation burden coefficient (within the model interface)
- ε : convergence tolerance governing merge decisions under uncertainty

A hierarchy of branching factor k over n agents has depth approximately $\log_k(n)$, representing the number of merge layers from leaves to the root interface.

4.1 Velocity decomposition

Net velocity is modeled as output contribution minus coordination drag and discard/evaluation burden:

- Output term: $v_{out}(n) = cn$
- Coordination drag term: $v_{coord}(n) = kn \log_k(n)$
- Discard/evaluation burden term: $v_{disc}(n) = mn\varepsilon^{-4}$

Yielding the model interface:

$$v(n) = cn - kn \log_k(n) - mn\varepsilon^{-4}.$$

4.2 Anti-precision clause

This model is a scope-bounded structural representation. It is not offered as a universal law, nor does it imply parameter fitting to any specific system.

5. Structural Results: Proved Lower Bound and Model Regime

Theorem 5.1 (Coordination Drag Lower Bound)

For all hierarchical coordination architectures in class $\mathcal{A}$ (satisfying nontrivial integration, bounded fan-in k , and per-item-per-layer cost floor), the total coordination cost satisfies

$$C_{\text{coord}}(n) \geq \alpha \cdot n \log_k(n) - O(n)$$

for some constant $\alpha > 0$.

Proof: The proof establishes that bounded fan-in forces average depth $\bar{d}(n) \geq \log_k(n) - O(1)$, and per-item-per-layer cost floor propagates this to total coordination cost. (Formal proof available in separate technical note.) $\square$

Implication: The coordination drag term $kn \log_k(n)$ in the velocity model is a structural inevitability, not an optimization artifact.

Model 5.2 (Discard Scaling Hypothesis)

Within the model interface, the discard/evaluation burden is modeled as

$$C_{\text{disc}}(n, \varepsilon) \sim m \cdot n \cdot \varepsilon^{-4}$$

where m captures expected evaluation cost and ε represents convergence tolerance.

Status: This scaling regime is a declared model assumption. The ε^{-4} dependence represents disproportionate evaluation cost as tolerance tightens, consistent with precision-amplified error propagation in hierarchical merge systems. Proving this as a structural lower bound (analogous to Theorem 5.1) requires formalizing ε -sensitivity propagation and coupling conditions—this is deferred to future work (see Note 1 below).

Implication: The discard term $mn\varepsilon^{-4}$ in the velocity model is a plausible scaling regime within the stated interface, but is not claimed as a structural inevitability.

Combined Velocity Form

Combining Theorem 5.1 (proved) and Model 5.2 (declared regime), the net velocity under the stated interface is

$$v(n) = cn - kn \log_k(n) - mn\epsilon^{-4}.$$

Treating n as continuous for a monotonicity check:

$$\frac{dv}{dn} = c - k(1 + \log_k(n)) - m\epsilon^{-4}.$$

Because $\log_k(n)$ grows without bound (Theorem 5.1), the coordination drag term $k(1 + \log_k(n))$ eventually exceeds any fixed constant. Combined with the modeled discard burden $m\epsilon^{-4}$ (Model 5.2), there exists n_0 such that $dv/dn < 0$ for all $n > n_0$.

Additionally, if $m\epsilon^{-4} > c$ (under Model 5.2), the discard/evaluation burden per agent exceeds per-agent output capacity even before coordination drag is counted; in this regime, beyond a finite scale, additional agents reduce net progress as measured by $v(n)$.

Note 1: The ϵ^{-4} exponent is consistent with observed behavior in precision-critical hierarchical systems, but the necessity of this specific power law requires proving that (a) validation boundaries are structurally forced, (b) rejection probability scales as ϵ^{-q} for some $q > 0$, and (c) the exponent $q = 4$ emerges from error coupling across hierarchy depth. This formalization is analogous to the Architecture Class proof for coordination drag (Theorem 5.1) and is left to future work.

6. Interpretation

6.1 Threshold meaning

The structural results (Theorem 5.1 + Model 5.2) do not claim that multi-agent systems “fail.” They establish that, within the specified architecture class, coordination drag (proved) grows unboundedly and discard/evaluation burden (modeled) scales disproportionately with precision, reducing net velocity beyond a threshold n_0 .

6.2 Negative velocity regime

When $m\varepsilon^{-4} > c$, the discard/evaluation burden per agent exceeds per-agent output capacity even before coordination drag is counted. In this regime, beyond a finite scale, additional agents reduce net progress as measured by the velocity function $v(n)$.

6.3 Why “marketing” language is admissible here

The opening market claim motivates the analysis (a public narrative of scalability). Theorem 5.1 establishes a structural constraint (coordination drag is inevitable), while Model 5.2 offers a plausible scaling regime (discard burden grows with precision requirements). The analysis constrains what is structurally possible within the specified architecture class.

7. Assumptions

These structural results are admissible only under the following core assumptions:

1. **Hierarchical coordination:** work is integrated through a hierarchy with branching factor k .
 2. **Partial discard:** merge interfaces discard some portion of candidate outputs to preserve coherence.
 3. **Uncertainty:** merge decisions are made under partial observability/noise and are governed by a tolerance ε .
 4. **Capacity definition:** c represents per-agent output capacity per coordination cycle within the model interface.
 5. **Discard/evaluation burden:** m captures expected evaluation/discard burden proportional to n within the model interface.
 6. **Depth approximation:** hierarchy depth is approximated by $\log_k(n)$ (balanced hierarchy assumption).
 7. **Declared scaling regime:** the ε^{-4} dependence is a declared regime representing disproportionate evaluation cost as tolerance tightens (not asserted as universal).
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8. Limits and Falsifiability

8.1 What would disconfirm this account (within scope)

This model-class account fails (requires revision) if, within the declared architecture class:

1. Coordination drag is shown to scale sub-logarithmically (disconfirming Theorem 5.1), **or** net velocity increases sustainably with n despite logarithmic coordination drag (disconfirming the combined model under Model 5.2's regime).
2. Tighter convergence tolerance ε does not increase evaluation burden in a way that can be represented by a super-linear scaling regime (the declared ε^{-4} or similar).
3. Discard can be shown to have no measurable coordination or legibility cost at scale within the same interface assumptions.

8.2 What does not disconfirm it

- Systems outside the declared architecture class (e.g., topologies without centralized merge discard)
 - Alternative modeling choices for the ε -dependence (this is explicitly declared)
 - Product-level benchmark results that optimize a different objective than “effective velocity” as defined here
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9. Implications

This paper does not recommend architectural responses. It identifies structural constraints that inform expectations:

- If an architecture must maintain coherence through hierarchical merging with partial output discard under uncertainty, then:
 - Coordination drag will grow unboundedly as $\Omega(n \log_k n)$ (Theorem 5.1, proved),
 - and if discard burden scales as modeled ($mn\varepsilon^{-4}$, Model 5.2), the combined effect will eventually dominate output gains as n grows (within the model interface).
 - Observed symptoms may include increased latency, repeated retries, rising cost-per-output, or degraded coherence—but those interpretations remain context-bound and are not offered as performance judgments.
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10. Provenance and Authorship

This work was authored by **James Roy Dennis**. All conceptual claims, interpretations, and conclusions expressed herein originate with the author, who bears full responsibility for the content.

AI-based tools may be used as editorial aids for clarity and organization. Such tools do not originate claims, arguments, or conceptual frameworks. Independent review, where present, functions as witness rather than collaborative authorship.

Appendix A — Notes Toward Formalization

This appendix outlines optional directions for formal strengthening. It is not required to sustain the architectural claim of this paper.

Possible extensions include:

- Discrete monotonicity analysis via finite differences
- Explicit bounding of $(n + 1) \log(n + 1) - n \log(n)$
- Regime case analysis for threshold behavior
- Continuous–discrete reconciliation

Future work may complete these derivations. Their absence does not alter the proof sketch presented above.